

- 3 (a) Fig. 3.1 is a photomicrograph of a transverse section through a region of the wall of the bronchus in the gas exchange system.

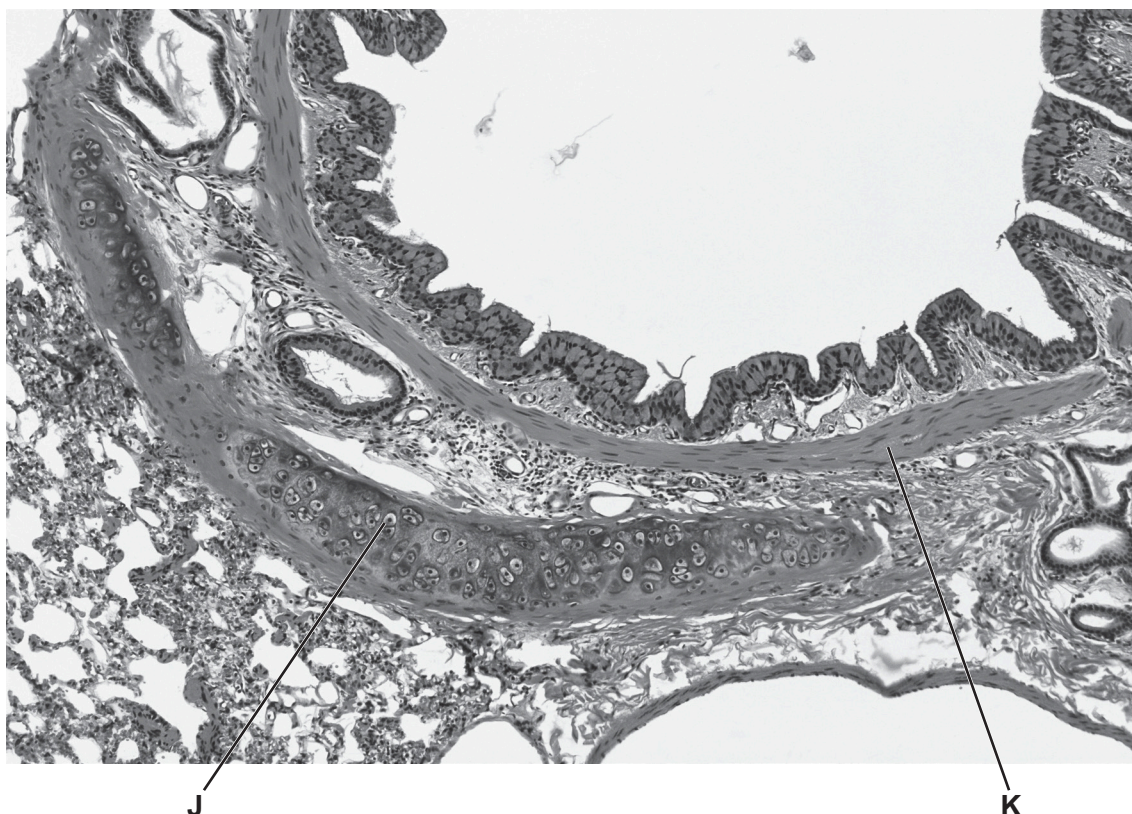


Fig. 3.1

Identify the tissues **J** and **K** shown in Fig. 3.1, **and** suggest how the wall of a bronchiole differs from the wall of the bronchus for these two tissues.

J

K

difference

.....

.....

.....

.....

.....

(b) Tuberculosis (TB) is an infectious disease that affects the human gas exchange system.

The pathogen that causes TB secretes a protein that can be detected in saliva.

Early diagnosis of TB is important in reducing the transmission of the pathogen.

Scientists have developed a test strip for TB that uses monoclonal antibodies. Monoclonal antibodies are specific in their action.

This test strip contains:

- mobile monoclonal antibodies that bind to one part of the protein secreted by the pathogen
- immobilised monoclonal antibodies.

Fig. 3.2 shows a simplified diagram of the test strip.

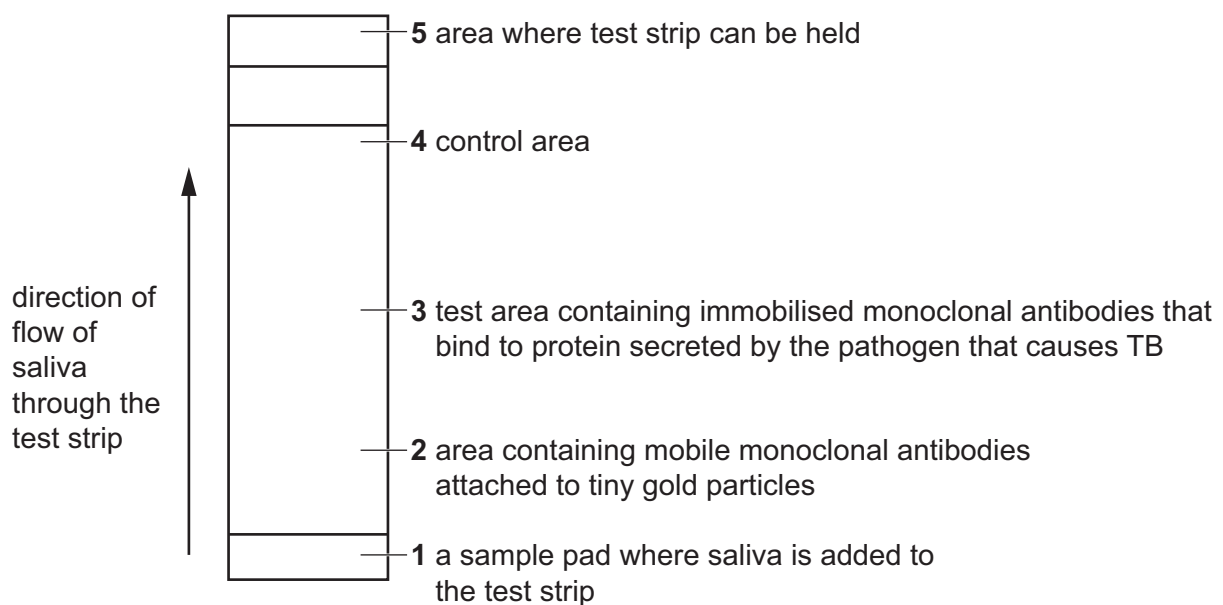


Fig. 3.2

A sample of saliva is collected and put onto the sample pad in the test strip.

The saliva moves up the test strip through area 2.

The mobile monoclonal antibodies are attached to tiny gold particles. If these antibodies collect in test area 3, a gold line becomes visible on the test strip.

A gold line that becomes visible in area 4 confirms that the test strip is working and that the results are valid.

(i) State the name of the pathogen that causes TB.

..... [1]

(ii) Name the part of the monoclonal antibody that binds to the protein from the pathogen.

..... [1]

- (iii) Saliva is added to a test strip to test for the presence of the protein secreted by the TB pathogen.

Fig. 3.3 is a diagram showing some of the molecules in area 3 of the test strip when a positive result for TB is obtained.

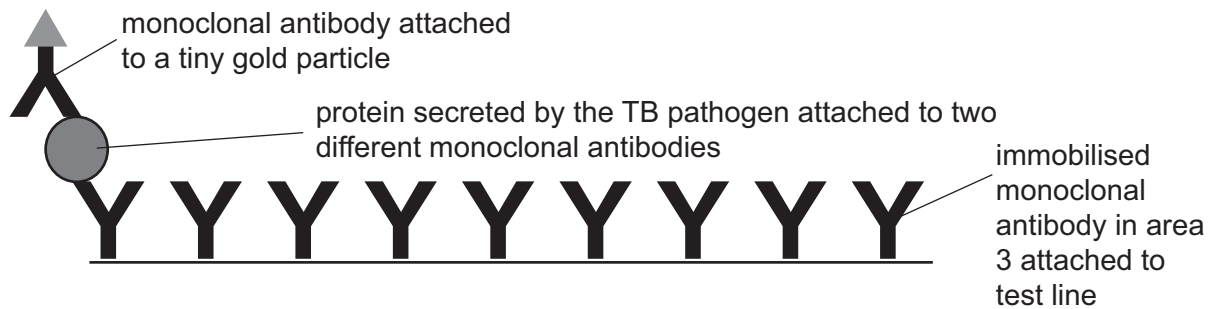


Fig. 3.3

Use the information in Fig. 3.3 to suggest **and** explain why this test is specific for TB.

.....

.....

.....

.....

.....

..... [2]

- (iv) Area 4 contains different immobilised antibodies to those in area 3.

The mobile monoclonal antibodies bound to tiny gold particles will bind to these immobilised monoclonal antibodies in area 4.

If the test has functioned correctly, a gold line will be visible in area 4.

Suggest how the structure of immobilised monoclonal antibodies in area 3 differs from the structure of the immobilised monoclonal antibodies in area 4.

.....

.....

.....

.....

..... [2]



- (c) Vaccination is another way of reducing the transmission of infectious diseases such as TB. The BCG vaccine is used to help control the spread of TB. This vaccine contains a weakened strain of the pathogen that causes TB. The BCG vaccine stimulates the development of antigen-specific memory T-lymphocytes.

Explain how memory T-lymphocytes provide protection from TB in a person who has been given a BCG vaccination.

.....

.....

.....

.....

.....

..... [3]

- (d) The bladder is the organ in the body used to store urine.

When cells divide uncontrollably in the bladder, a tumour develops. This can lead to bladder cancer.

The BCG vaccine has been used to treat bladder cancer.

The BCG vaccine is introduced into the bladder. The tumour cells take up the weakened pathogens in the vaccine and act as antigen-presenting cells.

- (i) Name the process used by the tumour cells to take up the weakened pathogens.

..... [1]

- (ii) Suggest how antigen presentation by tumour cells stimulates an immune response that leads to the destruction of the tumour cells.

.....

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 16]

